

Hussein Shalaby

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EDUCATION

University of Padua

Bachelor of Science in Information Engineering

Padua, Italy

Expected Feb. 2027

- Relevant coursework: Computer Architecture, Algorithms & Data Structures, Linear Algebra, Machine Learning, Signals & Systems, Probability Theory.

PROJECTS

Renderer | *C++*, *OpenGL*, *Vulkan* | [github](#)

- Built a C++20/OpenGL renderer with model loading, PBR shading, and image-based lighting.
- Implemented anti-aliasing, HDR, bloom effects and real-time shadows for directional, spot, and point lights.
- Added Vulkan initialization with OpenGL fallback and a Dear ImGui debugging interface.

CUDA Edge Detection | *CUDA*, *C++* | [github](#)

- Implemented a Sobel edge detector in CUDA, validated bit-for-bit against a CPU reference before benchmarking.
- Profiled with Nsight Compute to show shared-memory tiling ran 1.8x slower than the naive kernel.
- Identified uncoalesced global loads as the true bottleneck rather than global memory traffic.

Software Rasterizer | *Rust* | [github](#)

- Built a software rasterizer from scratch in Rust utilizing the CPU only.
- Rasterizes obj models through a MVP and viewport pipeline, using barycentric coordinates with a z-buffer for depth and back-face culling.
- Samples tga textures from interpolated UVs, includes a look-at camera and obj loader.

3D Gaussian Splatting Renderer | *Rust*, *wgpu*, *rust-gpu* | [github](#)

- Built a real-time 3D Gaussian Splatting renderer in Rust using wgpu and splat sorting on the CPU.
- Wrote GPU shaders in Rust using rust-gpu

OPEN SOURCE

Filament (Google) | *C++* – [contribution](#)

- Added `to_string()` conversions for public enums in Google's Filament real-time rendering engine.

ACHIEVEMENTS

ICPC SWERC 2024 Finalist | *Southwestern Europe Regional*

Pisa, Italy

- Selected to represent the University of Padova against 180+ teams from across southwestern Europe.

Italian Collegiate Programming Contest (ITACPC)

Italy

- Top 20 nationally (2025) and top 30 (2024) among 200+ university teams.

MATE ROV Regional – 4th place | *Underwater Robotics*

Alexandria, Egypt

- Wrote Python/OpenCV image-processing for the vehicle's autonomous missions.

EXPERIENCE

Freelance Software Developer

Aug. 2025 – Sep. 2025

Self-Employed

Remote

- Cut active-window detection latency from 5,000ms to under 100ms by using native platform event hooks.
- Built a parametric CAD tool in Python to automate pressure-vessel layout, saving clients hours of manual work.

IT Systems Intern

Aug. 2023 – Sep. 2023

Intercom Enterprises

Cairo, Egypt

- Completed Dell ISMv5, Enterprise Cloud, and RHEL training in datacenter virtualization and storage.
- Reviewed rack storage configurations for load balancing during an enterprise disaster-recovery assessment.

TECHNICAL SKILLS

Languages: C++, C, Rust, Python, Go, SQL

Development: Git, CMake, GDB, Docker, Linux

Graphics & GPU: OpenGL, Vulkan, CUDA, GLSL, RenderDoc, Nsight